LLMs x Music: Scalable Language Modeling Unlocks Music Applications

Zachary Novack
Spotify
znovack@ucsd.edu

Seungheon Doh
Sony Group Corporation
seungheon.doh@gmail.com

Huan Zhang
Clefer
huan.zhang@qmul.ac.uk

Yewon Kim
Carnegie Mellon University
yewon@cmu.edu

Sang-gil Lee
Nvidia
tkdrlf9202@gmail.com

1 Motivation and Relevance

Large Language Models (LLMs) have fundamentally transformed how machines understand and generate human language, emerging as foundational models that exhibit remarkable capabilities across diverse domains. To harness these models effectively, a deeper understanding of their emergent properties is required: massive instruction following gives rise to multi-task generalization [3], chain-of-thought reasoning unlocks structured problem solving [25], preference alignment through post-training shapes model behavior toward human intent [16], and tool calling enables agentic workflows by facilitating interaction with external environments [18]. Building on this foundation, this tutorial is motivated by the need to go beyond a surface-level introduction to LLMs and product API usage. Instead, it explores the evolving trajectory of large-scale models and their practical usage, reviewing the specific properties that make them powerful for music understanding, recommendation, co-creation, and education.

Moving from theoretical properties to technical implementation, we examine how these emergent properties push music understanding well beyond simple classification and captioning models (Chapter 2). Where earlier models were largely confined to single-task, single-modality pipelines, instruction following and multi-task generalization enable a single LLM to unify captioning, question answering, and reasoning across audio [10, 11], symbolic notation [30], and textual metadata [17] within one framework. We also discuss the current limitations of LLMs in musical understanding and outline future directions for building more capable and musically grounded systems [22, 31].

Ultimately, the core of this tutorial lies in the practical applications of LLMs in music (Chapter 3). We examine how LLMs enable new paradigms for music listeners through conversational recommendation and retrieval systems [6, 7], and empower music producers with intelligent co-creation tools for composition [30, 21], post-production [8, 4], and collaborative song-writing [9]. Furthermore, we investigate how LLMs support performers & educators with feedback systems [32] and enhanced music notation understanding [33]. Throughout, we emphasize the multimodal nature of modern LLM systems and their potential to bridge audio, symbolic, and textual modalities in music applications.

We believe this tutorial will provide substantial value to the MIR community by offering both theoretical foundations and practical insights into leveraging LLMs for music. Participants will gain hands-on experience through code practices, access to relevant benchmarks and evaluation methodologies, and a comprehensive understanding of the current state and future directions of LLM-based music systems.

2 Content Overview

Chapter 0. Introduction (5 min)

This chapter introduces the tutorial, outlining the motivation for exploring LLMs in music applications and presenting the tutorial structure. We provide an overview of how LLMs are transforming MIR research and applications.

Chapter 1. Overview of Large Language Models (30 min)

This chapter provides a comprehensive foundation on LLMs, explaining why they are considered foundational models and how they achieve their remarkable capabilities. We cover the technical underpinnings from autoregressive modeling to advanced post-training techniques.

- Training Objectives: Pre-training, Mid-training, Post-training [19, 16]
- Advanced Capabilities [24]
  - In-Context Few-shot Learning [29]
  - Conversationability and Multi-turn Dialogue
  - Instruction Following [23, 3]
  - Chain-of-Thought Reasoning [25, 28]
  - Tool Calling [18]
- Agentic Pipelines

Chapter 2. LLMs for Music Understanding (30 min)

This chapter explores how LLMs are being applied to understand music across various tasks and representations. We examine the challenges of connecting language with musical content and survey state-of-the-art approaches for music captioning, question answering, and reasoning.

- Music Perception, Understanding, and Reasoning
- Large Music-Language Models
  - Model Design (Encoders, Adapters, Decoders)
  - Audio [10, 11]
  - Score Images [20, 1]
  - Symbolic Text Notation [30]
  - Text Metadata [17]
- Benchmarks and Evaluation [22, 31, 15]

Chapter 3. LLMs for Musical Applications (120 min)

This chapter presents practical applications of LLMs across the music ecosystem, covering three main user perspectives: listeners, producers, and performers. We explore both audio and symbolic modalities for each application domain.

LLMs for Listeners: Annotation, Retrieval and Recommendation

- Music Annotation and Descriptions [10, 11]
- Conversational Music Recommendation [6, 7]

LLMs for Producers: Composition and Post-Production

- Human-AI Collaborative Songwriting [13, 27]
- Composition Support Tools [5, 21, 26]
- Conversational Post-Production [4, 8]

LLMs for Performers: Education and Feedback

- Multi-modal understanding performances [33, 2]
- Assessment and Coaching [32, 12, 14]

- Interactive Music Education with LLMs

Chapter 4. Conclusion

- Recap
- Future Research Directions
- Q&A and Discussion

3 Originality and Novelty

Large language models are now ubiquitous—no longer confined to natural language processing, they have become a unifying backbone across vision, audio, and speech domain, and it is safe to assume that ISMIR attendees in 2026 already encounter LLM-based systems well beyond text. Against this backdrop, the relevant question is no longer *what is an LLM* but *which properties of LLMs actually matter for music*, and *how do we harness them in practice*. This framing places the proposed tutorial in distinct territory from prior work.

While the tutorial shares thematic roots with ISMIR 2023 [1] (Transformer-based Symbolic Generation) and ISMIR 2024 [2] (Connecting Audio and Natural Language), those sessions were grounded in the conventional language modeling paradigm—next-token prediction $p(x_t|x_{<t})$ and task-specific cross-modal alignment with moderately sized models (GPT-2, BERT, T5)—and predate the scaling-driven transition to foundation models. Even among LLM-focused tutorials, the common framing is applicative: *use an LLM for task X*. What is missing is an explicit treatment of *why* certain tasks become tractable at scale, and *how* specific emergent capabilities—in-context learning, chain-of-thought reasoning, tool calling, and instruction following—can be deliberately engineered into musical pipelines.

This tutorial fills that gap. Rather than surveying LLMs for music in general, we take an emergent-capability-first perspective: each chapter is organized around a capability class, grounding it in the underlying mechanism before demonstrating musical applications. Participants leave not just knowing *that* LLMs can annotate, compose, or coach, but understanding *which lever to pull* and *why it works*—knowledge that transfers to new tasks as the model landscape continues to evolve.

4 Presenters

Zachary Novack

Zachary Novack is a Research Scientist at Spotify in the Artist First AI-Music Lab. Graduating in 2026 with a PhD from UC San Diego, Zachary's research has focused both on efficient interactive music generation and large scale evaluation and benchmarking for LLMs for musical reasoning. He has received numerous awards across diverse venues, including oral presentations at ICML, ICASSP, and Interspeech, spotlight papers at ICLR and NeurIPS, and recently Best Paper at ISMIR 2025 for work on diagnosing Music-LLM evaluation. He co-presented the tutorial "Connecting Music and Natural Language" in ISMIR 2024 and organized the AI Music workshop at NeurIPS 2025. Outside of academics, Zachary is passionate about music education and teaches competitive indoor drumline in the southern California area.

Seungheon Doh

Seungheon Doh is a Research Scientist at Sony. His research focuses on conversational music understanding, recommendation, and creation. He has published research on large language models for music applications at venues including ISMIR, ICLR, IEEE ICASSP, and IEEE TASLP. He co-presented a tutorial on "Connecting Music Audio and Natural Language" at ISMIR 2024 and has organized several workshops (NLP4Musa, LLM4MA) exploring the intersection of music and language. Seungheon has interned at Sony AI, Adobe Research, Chartmetric, NAVER, and ByteDance.

Huan Zhang

[1] https://docs.google.com/presentation/d/1AB27cUQzXaKbZlh3xtOCTWHR-PdWWXrF8K9xROFoNvk

[2] https://mulab-mir.github.io/music-language-tutorial/intro.html

Huan Zhang is a Research Scientist at Clefer Inc., exploring how LLMs can understand, generate, and evaluate performance expression in music. She has developed multimodal models such as LLM coach that provides formative feedback on performances, and framework for text-controlled expressive performance generation. Huan has collaborated with industry leaders such as Sony CSL and Yamaha, and is deeply committed to applying her research to music education. Huan also serve as the MIREX Task Captain of RenCon (Expressive Performance Rendering Competition) and Music Performance Difficulty.

Yewon Kim

Yewon Kim is a Ph.D. student in the Computer Science Department at Carnegie Mellon University, where she is advised by Chris Donahue. Yewon's research explores how generative AI can enhance human creativity in music, developing interactive systems and methods for human-AI music co-creation. She has published across HCI (ACM CHI, ACM DIS) and AI (NeurIPS) venues, and was recognized with a Best Paper Award at ACM CHI. She has also interned at Adobe Research.

Sang-gil Lee

Sang-gil Lee is a Research Scientist on the Applied Deep Learning Research team at NVIDIA, where his research focuses on audio AI across sound, music, and speech. He received his PhD in Electrical and Computer Engineering from Seoul National University. His research spans neural waveform synthesis, audio codecs, audio generation, and multimodal audio-language models. He has contributed to a number of influential audio AI systems, including BigVGAN, ETTA, Audio Flamingo 3, Music Flamingo, UALM, and Nemotron-Labs-Audex. His recent work explores unified audio-language models that jointly support understanding, generation, and reasoning across sound, music, and speech.

5 Intended and expected audience

- Researchers who are interested in LLMs and music domain-specific finetune methods
- Students and supervisors who want to discover musical appclaitions with LLMs

6 Presentation tools and format

- Presentation Slide (Pdf)
- Web Book (Jupyer Book)
- Reproducible code (GitHub)

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